Jake O'Reilly

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Education

BSc in Computer Science — Expected May 2028
Technological University Dublin — *Dublin, Ireland*
- **Relevant Coursework:** Data Structures & Algorithms, Generative AI Programming, Databases, Data Analysis

Experience

Software Engineer Intern — Jun 2026 - Present
Speed-Deed — *Dublin, Ireland*
- Cut per-dashboard query volume ∼95% (19 live queries to one snapshot read) by architecting a cron-driven PostHog/PostgreSQL/Recharts pipeline that pre-computes funnel, timing, and cost metrics into a single row
- Restored a production upload failure affecting live pilot users in under 10 minutes by tracing 429 errors to a database migration that had not been applied to production and running it against the live instance mid-pilot
- Gave admins visibility into third-party spend across 4 categories by building an immutable cost-events ledger that logs each priced action at a frozen FX rate and survives property deletion
- Closed 4 authorization and data-integrity gaps by locking sensitive write endpoints to the owning user and making retried webhooks and reopened tasks apply their effect only once, preventing duplicate charges and corrupted task state

Founder Programme Participant — May 2026
Hatch105 — *Dublin, Ireland*
- Selected as 1 of 27 (top 1.5% from 1,700+) applicants for Hatch105's inaugural founder programme, pitching a technical AI prototype to a panel of founders and CEOs
- Built ClearStep, an AI-powered SaaS that replaces manual store-data analysis with a real-time dashboard of prioritised business actions for merchants without dedicated data teams
- Engineered an LLM pipeline in TypeScript using the Claude API to process raw Shopify data into structured JSON, with custom prompt logic for priority ranking

Software Engineering Insight Programme — Apr 2026
Bank of America — *Dublin, Ireland*
- Awarded 1st of 4 teams (16 participants) for identifying and patching critical authentication vulnerabilities and delivering a Savings Goals feature, judged by Bank of America engineers
- Eliminated session-hijacking risk by replacing plain-text password storage with bcrypt hashing and implementing a JWT blacklist in Node.js/PostgreSQL to block post-logout token reuse
- Built 5 REST endpoints from scratch for the Savings Goals feature using Zod validation and Drizzle ORM migrations, covering the full goal lifecycle: creation, deposits, deletion, and completion

Projects

ThreadBase | GitHub | *Python, FastAPI, PostgreSQL, Docker, GitHub Actions*
- Built a GitHub Actions CI/CD pipeline that runs a 20+ case Pytest suite against a Dockerised Postgres container, auto-deploying to Heroku only after tests pass
- Built a RESTful forum API secured with OAuth2 password-flow, JWT bearer tokens, and bcrypt hashing, supporting full CRUD for authenticated posting and voting with ownership enforcement
- Designed a normalised 3-table PostgreSQL schema with SQLAlchemy ORM and Alembic migrations, enforcing referential integrity through foreign-key constraints with cascading deletes

HushPath | **3rd Place, Workday Hackathon** | *Python, REST APIs*
- Built HushPath, a sensory-friendly journey planner, scoring Dublin city centre across 80m grids using the Overpass and Nominatim APIs to calculate POI density as a noise proxy
- Generated noise-aware walking routes between any two points by integrating the Google Maps API with a custom waypoint-selection algorithm prioritising lower-density grid cells over the shortest path

Reservr | GitHub | *PHP, MySQL*
- Built a multi-user library reservation system, preventing duplicate and conflicting bookings through a normalised 4-table schema and strict server-side validation in PHP/MySQL
- Implemented role-scoped access control and session-based authentication to manage concurrent reservations across mobile and desktop viewports

Technical Skills

Languages: Python, SQL, TypeScript, JavaScript, Java, PHP, C, HTML, CSS
Frameworks & Libraries: FastAPI, Node.js, SQLAlchemy, Pytest, Tailwind CSS
Databases: PostgreSQL, MySQL, Supabase
Developer Tools & Platforms: Git, GitHub Actions, Docker, AWS, Google Cloud, Cloudflare, Linux, Vercel, Heroku